

Theorem [Villani '09; 6.9]:

$$\alpha_k \rightharpoonup \beta \iff W_p(\alpha_k, \beta) \rightarrow 0$$

aka W_p metrizes the weak* convergence

Definition [weak* topology]:

We say that $(\alpha_k)_k$ converges weakly* to $\alpha \in \mathcal{P}(\mathcal{X})$ (denoted as $\alpha_k \rightharpoonup \alpha$) iff for

any continuous $f \in \mathcal{C}(\mathcal{X})$,
$$\int_{\mathcal{X}} f \, d\alpha_k \rightarrow \int_{\mathcal{X}} f \, d\alpha$$

Corollary [Villani '09; 6.11]:

$$\alpha_k \rightarrow \alpha, \beta_k \rightarrow \beta \Rightarrow W_p(\alpha_k, \beta_k) \rightarrow W_p(\alpha, \beta)$$

Significance for ML

- W_p is finite + meaningful even when supports don't overlap ($\neq$ KL, TV)
- Stability: small perturbations in data
→ small W_p
- Empirical distributions converge to the data-generating law in W_p

Weak Convergence and W_p

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Why geometry awareness matters

$$\text{TV}(\delta_{x_t}, \delta_x) = 1$$

$$\text{KL}(\delta_{x_t}, \delta_x) = \infty$$

$$W_p(\delta_{x_t}, \delta_x) = d(x_t, x)$$

